Scientific Report: Plant Acoustic Emissions

Experiment Start: 2026-06-24 01:46:15
Last Update: 2026-06-24 01:46:15
Duration: 0:00:00.000008
Session: 20260624_010136

Baseline Established: No
Baseline Duration: 0.0 minutes

METHODOLOGY

Statistical Methods:
• Significance Level: α = 0.05
• Drift Detection: ±2σ threshold
• Minimum Occurrences: 5 detections
• Frequency Binning: 1 kHz bins
• Trend Analysis: Linear regression

Tracking Metrics:
• Channel RMS and Peak amplitudes
• Persistent frequency peaks
• Frequency drift detection
• Soil moisture correlation
• Event rate analysis

SUMMARY STATISTICS

Channels Tracked: 0
Persistent Peaks: 0
Total Events: 0
Event Types: 0

Significant Changes Detected: 0

Experimental Configuration

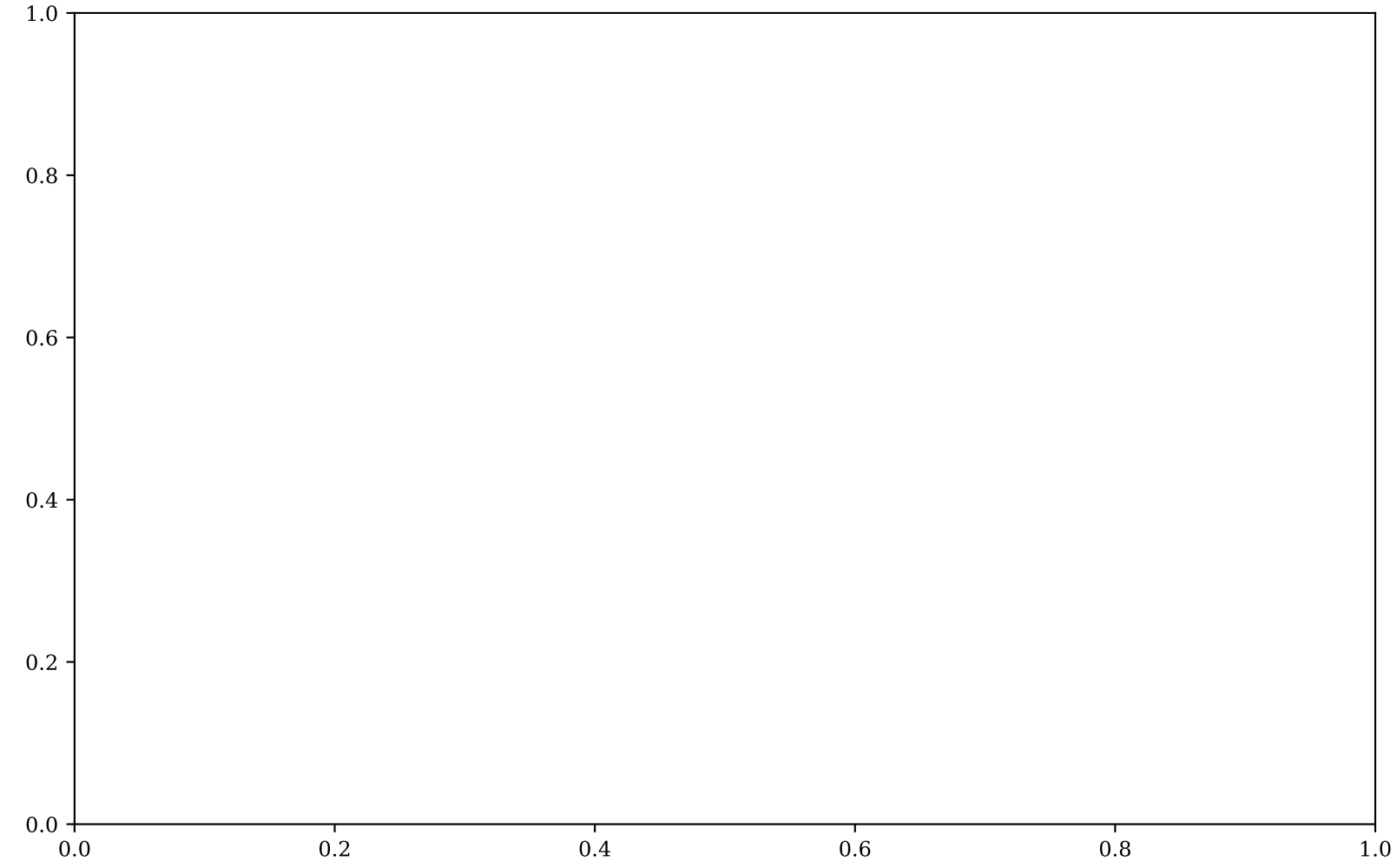
OSCILLOSCOPE CONFIGURATION (Updated 2026-06-22)

Acquisition Profile: deep_memory_500k
Sample Rate: 500 kHz
Memory Depth: 300,000 points
Chunk Points: 250,000 points
Max Frequency: 100 kHz
Nyquist Frequency: 250 kHz

Time Base: 10 ms/div
Trigger Mode: EDGE
Trigger Source: CHAN1
Trigger Level: 20 mV

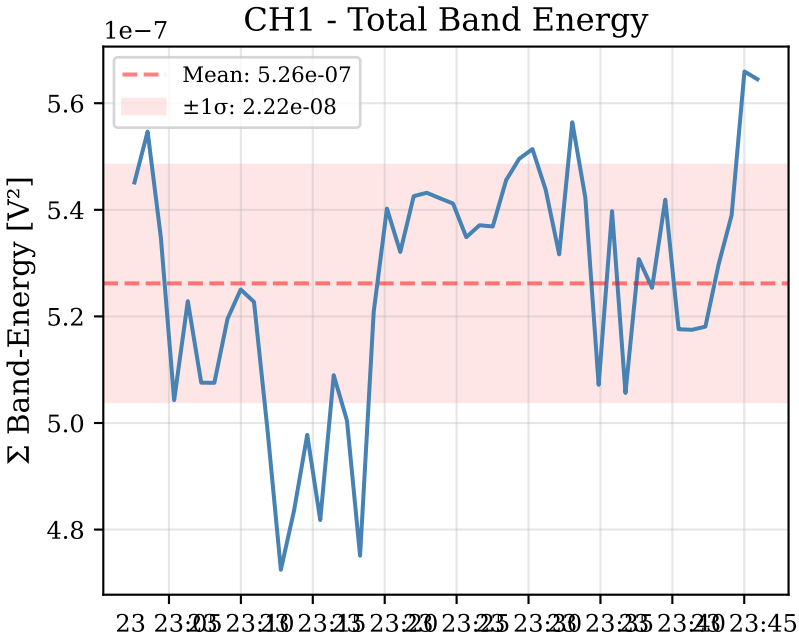
Channels:
• CH1: Piezo + LM358 amplifier + 820 kΩ (10:1 probe, passive, EM reference)
• CH3: Piezo + amplifier + 820 kΩ (10:1 probe, soil near plant)
• CH4: Piezo + amplifier + 820 kΩ (10:1 probe, stainless-steel rod next to plant)
• CH2: DISABLED (hardware fault, replaced by CH4 on 2026-06-22)

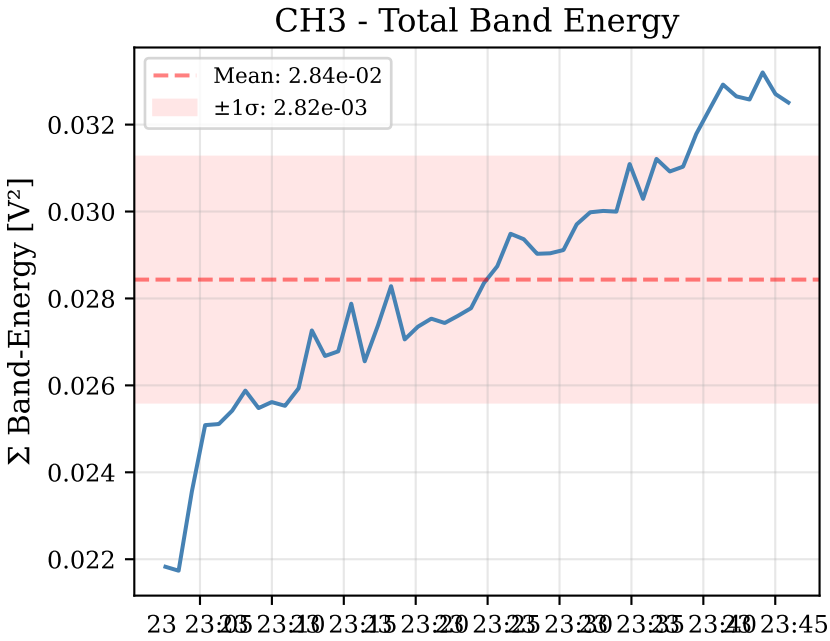
TECHNICAL IMPROVEMENTS:
• Reduced sample rate: 25 MSa/s → 500 kHz
• Increased acquisition window: 12 ms → 600 ms
• Focused on biologically relevant range: 0-100 kHz
• Improved stability for long-running experiments

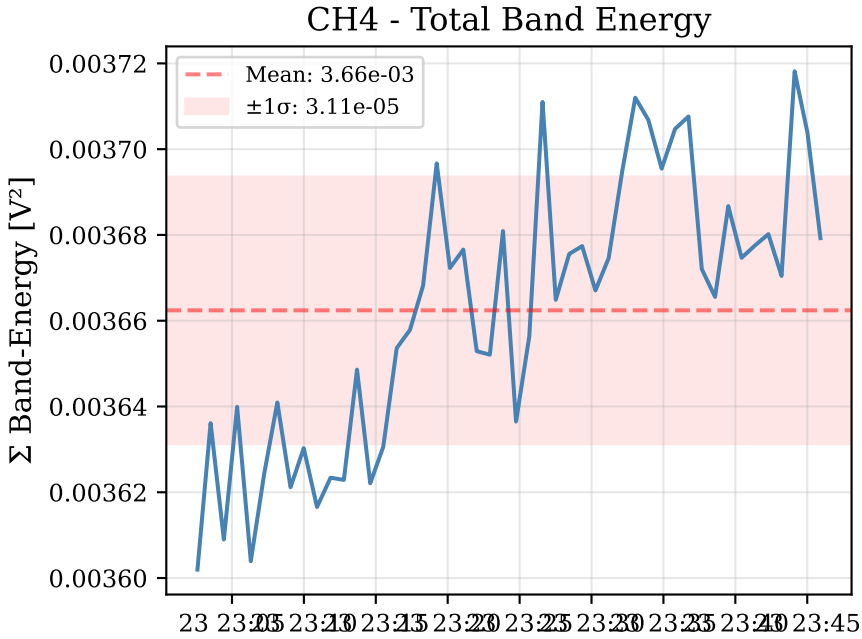


Channel Statistics (Phase 2: CH1+CH3+CH4, CH2 disabled)

Channel Statistics Summary

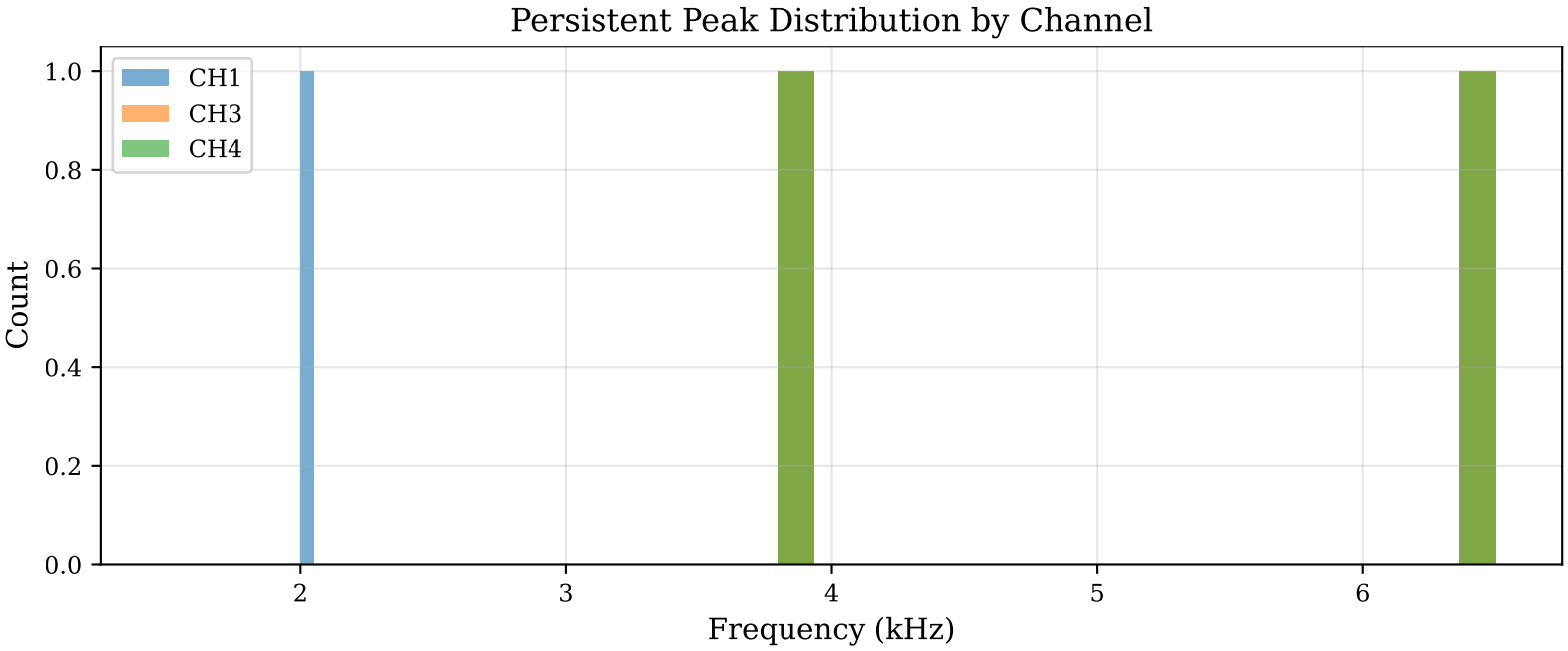


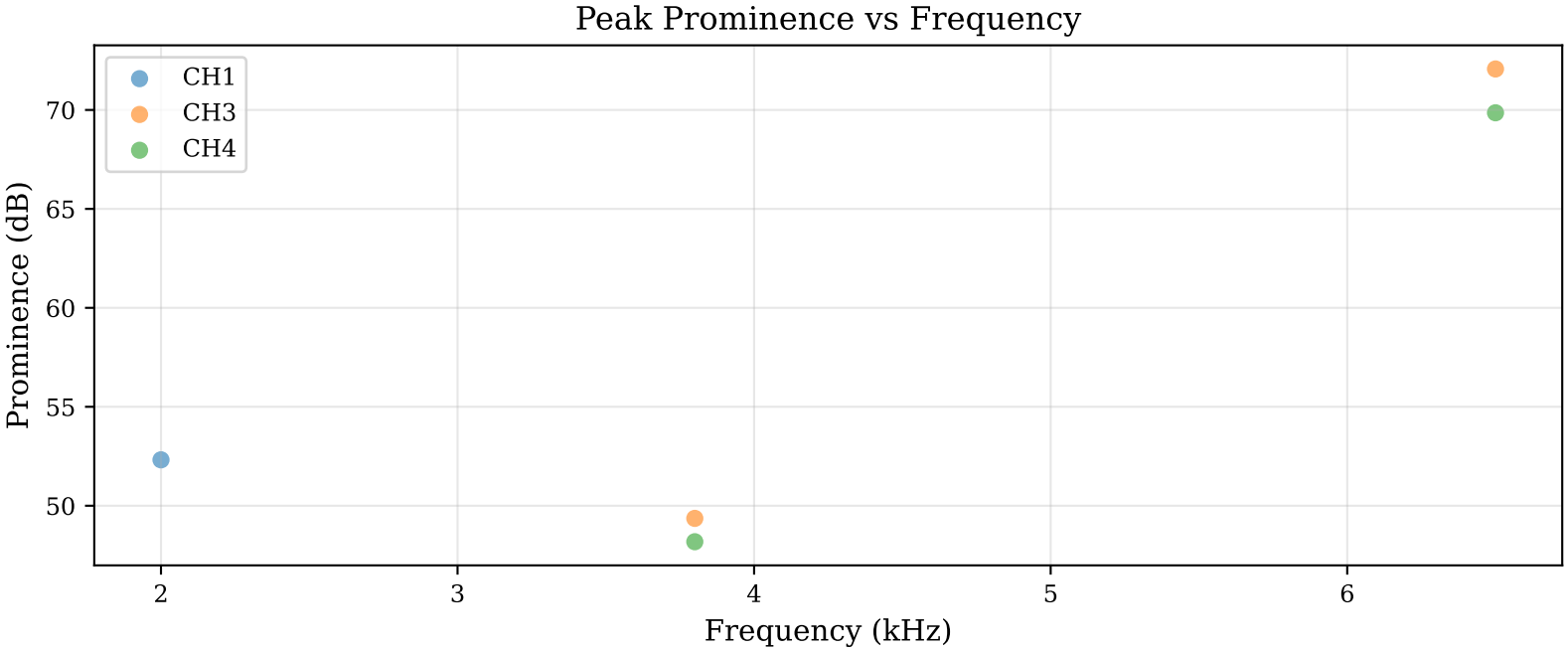




Channe	Mean [V²]	Std	Min	Max	N
CH1	5.26e-07	2.22e-08	4.72e-07	5.66e-07	48
CH3	2.84e-02	2.82e-03	2.17e-02	3.32e-02	48
CH4	3.66e-03	3.11e-05	3.60e-03	3.72e-03	48

Persistent Frequency Peaks





Top 10 Persistent Peaks by Prominence (rekonstruiert aus frame['peaks']: ≥50% presence, ≥40 d

CH	Freq (kHz)	Prom (dB)	Occ	%Frames	Trend
CH3	6.5	72.1	48	100	stable
CH4	6.5	69.9	48	100	stable
CH1	2.0	52.3	35	73	stable
CH3	3.8	49.4	40	83	stable
CH4	3.8	48.2	44	92	stable

Frequency Distribution Analysis

No frequency detections available

Soil Moisture Analysis

No soil moisture data available

Summary and Conclusions

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KEY FINDINGS (rekonstruiert aus frame['peaks'])

1. Persistent Frequency Peaks
   • Total persistent peaks: 5
   • CH1: 1 peaks
   • CH3: 2 peaks
   • CH4: 2 peaks
   • CH2: DISABLED (hardware fault)

2. 39 kHz Region Analysis
   • Detections in 38-40 kHz range: 0
   • Channels affected: none
   • Mean prominence: nan dB

3. Frequency Drifts Detected
   • Peaks with significant drift: 0
   • Drift criteria: ≥12 points, ≥500 Hz displacement, R²≥0.8

4. Channel Activity
   • Most active channel: CH3
   • Least active channel: CH1

TECHNICAL IMPROVEMENTS (2026-06-22)

• Sample Rate: 25 MSa/s → 500 kHz
• Acquisition Window: 12 ms → 600 ms
• Memory Depth: AUTO → 300,000 points
• Frequency Resolution: 250 Hz bins (was 1 kHz)
• Drift Detection: Enhanced with R² threshold

CONCLUSIONS

The experiment has identified 5 persistent frequency peaks
across all three channels. The technical improvements enable longer acquisition
windows and more precise frequency tracking.

The 39 kHz region shows some
activity, particularly on CH3 and CH4 (Phase 2 setup; CH2 disabled due to hardware fault, replaced by CH4 o

RECOMMENDATIONS

• Continue monitoring for trend analysis
• Investigate 39 kHz activity for biological significance
• Correlate with environmental parameters
• Consider additional sensors for validation

STATISTICAL CONFIDENCE

• Significance level: α = 0.05
• Minimum occurrences: 5 detections
• Frequency resolution: 250 Hz bins
• Tracking duration: 0.0 minutes
• Drift detection: R² ≥ 0.8, slope ≥ 0.5 Hz/s
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